40. Noncommutative Algebras

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The main theorems of commutative algebra are, as is known, contained in Galois theory, preceded by the theory of adjunction fields and splitting fields: fields sufficient for a given polynomial to split off one linear factor, or to split completely into linear factors.

Here I develop the corresponding parts of algebra in the noncommutative case, especially in the hypercomplex case. In principle I work with noncommutative methods, namely representation in noncommutative fields. At the end I show how the above-mentioned theorems of commutative algebra can be proved in a completely parallel manner by means of representation in commutative fields.

The underlying representation theory, preceded by a short automorphism theory (§1), is a further development of the theory based on representation modules. In particular, alongside direct representation I consider reciprocal representation. Each can be reduced to the other, but the reciprocal representation is now generated by the reciprocal representation module. The advantage is that the reciprocal representation module, hence a bimodule, can also be regarded as a one-sided module over an extension ring (§2). Thus representation in noncommutative fields is reduced to ideal theory in the extension ring; the irreducible representation classes correspond to the irreducible ideal classes of the extension ring, in exact analogy with the facts which hold for a hypercomplex system itself when represented over its commutative coefficient field (§§3, 4).

From this point one obtains the structural theorems for matrix rings over noncommutative fields by observing (§5) that every subring gives a representation by this field, or equivalently a reciprocal representation by the reciprocally isomorphic field. This reciprocally isomorphic field is a first analogue of a minimal adjunction field and splitting field in the commutative case: it mediates all reciprocal representations of first degree, and, when the rank over the center is finite, also a full decomposition into direct summands of rank 1. This gives the Galois theory for fields (§6) and also expresses the fact (§7) that reciprocally isomorphic division algebras, i.e. fields of finite rank over their centers, generate inverse classes in R. Brauer's group of algebra classes.

A second proof of the Galois theory, valid more generally for simple systems, is almost an immediate consequence of a theorem on commuting subrings, which in turn is connected almost directly with the preceding automorphism considerations. Up to this point the arguments are purely noncommutative. But the question of commutative splitting fields is also treated noncommutatively by representing the splitting field through the division algebra, using the observation prepared in §5 (§7). The final part gives the transfer to the commutative case (§8) and the theory of splitting fields for arbitrary systems (§9).

R. Brauer based the theory of splitting fields on this representation theory in the commutative case, and on irrational factor systems. Because of this commutative foundation he, and later Albert, had to assume the center to be a perfect field, a restriction which is unnecessary in the noncommutative foundation. With the same restriction, again using representation theory in the commutative case, R. Brauer and K. Shoda further developed the theory.

Finally I mention that, for fields, a short account is found in van der Waerden, following my lecture course of summer 1928. Van der Waerden's account introduced several simplifications relative to that course; some of these I adopted in a second course, and some only here. The theorem on commuting subrings (§5) and the second proof of the noncommutative Galois theory based on it (§6, 1 and 2) were added only later. Certain arguments of the first course, the method of forming intersections of difference ideals, have the advantage, though they are more complicated, of transferring to systems of infinite rank and to integral systems. For commutative integral systems this leads to the connection between ideal differentiation and the different.

The principal application of the theory of splitting fields lies in the theory of crossed products and their factor systems, which in turn gives the basis for applications in number theory; this will not be pursued here.

§1. On Automorphisms, Modules, and Bimodules

Representation theory based on representation modules rests on the theory of the automorphism ring of abelian groups, with or without operators. The implicit arguments involved, namely the relations between mappings and calculation laws, are formulated here as a few simple propositions in order to avoid repetitions.

1. Multiplicative mapping. Associative law. Let $\mathfrak{G}$ first be a group without operators, and let $\mathfrak{A}$ be its absolute automorphism domain, i.e. the system of all homomorphisms of $\mathfrak{G}$ into itself. $\mathfrak{A}$ is closed under multiplication, since the product $\sigma\tau$ is defined by

$$g(\sigma\tau) = (g\sigma)\tau, \qquad g \in \mathfrak{G}, \quad \sigma, \tau \in \mathfrak{A}, \tag{1}$$

and plainly satisfies the associative law.

The group $\mathfrak{G}$ becomes a group with operators when a set $\mathfrak{B}$ of symbols $O, H, \dots$ is given so that the products $gO, gH, \dots$ are well-defined elements of $\mathfrak{G}$ and generate homomorphisms of $\mathfrak{G}$ into itself:

$$(gh)O = gO \cdot hO.$$

If the operator domain $\mathfrak{B}$ is multiplicatively closed, then the induced map into the automorphism domain is multiplicatively homomorphic if and only if the associative relation

$$g(OH) = (gO)H, \qquad g \in \mathfrak{G},\ O, H \in \mathfrak{B}, \tag{1a}$$

is satisfied. Correspondingly, a reciprocal homomorphism is obtained for left operators.

2. Operator-homomorphic mapping. If $\mathfrak{G}$ is a group with operators, operator-homomorphy is defined by

$$(gO)\sigma = (g\sigma)O, \tag{2}$$

and, for left operators, by

$$(Og)\sigma = O(g\sigma). \tag{2*}$$

For two operator domains $\mathfrak{B}$ and $\mathfrak{C}$, the elements of $\mathfrak{C}$ generate $\mathfrak{B}$-automorphisms, and at the same time the elements of $\mathfrak{B}$ generate $\mathfrak{C}$-automorphisms, if and only if the domains are commutatively connected with $\mathfrak{G}$, respectively if the running associative law is fulfilled:

$$(gO)H = (gH)O, \tag{2a}$$

respectively

$$(OH)g = O(Hg). \tag{2a*}$$

3. Modules and bimodules over rings. A right module $\mathfrak{M}$ over a ring $\mathfrak{R}$ is an additive abelian group with the elements of $\mathfrak{R}$ as right operators, satisfying the associativity relation and the distributive relations

$$(g + h)\rho = g\rho + h\rho, \qquad g(\rho + \sigma) = g\rho + g\sigma. \tag{3a}$$

The corresponding definitions hold for left modules.

There are two kinds of bimodules. Right modules over two rings $\mathfrak{R}$ and $\mathfrak{S}$ are called bimodules when, in addition to the individual module laws, one has

$$(m\rho)\sigma = (m\sigma)\rho.$$

An $\mathfrak{R}$-left, $\mathfrak{S}$-right module is a bimodule when the running associative law

$$\rho(m\sigma) = (\rho m)\sigma$$

is added.

If the operator domain $\mathfrak{R}$ of an additively written abelian group $\mathfrak{M}$ is a ring, then the map from $\mathfrak{R}$ into the absolute automorphism ring is ring-homomorphic precisely when $\mathfrak{M}$ is a right $\mathfrak{R}$-module; for left modules it is reciprocally ring-homomorphic. If $\mathfrak{M}$ is a right module over $\mathfrak{R}, \mathfrak{S}$, then $\mathfrak{M}$ is a bimodule precisely when $\mathfrak{R}$ maps homomorphically into a subring of the $\mathfrak{S}$-automorphism ring and $\mathfrak{S}$ maps homomorphically into a subring of the $\mathfrak{R}$-automorphism ring. For $\mathfrak{R}$-left, $\mathfrak{S}$-right bimodules, the $\mathfrak{R}$-map is reciprocal and the $\mathfrak{S}$-map direct.

In particular:

1) If $\mathfrak{R}$ is a ring with identity, then $\mathfrak{R}$ is directly isomorphic to its automorphism ring as a left $\mathfrak{R}$-module, and reciprocally isomorphic to its automorphism ring as a right $\mathfrak{R}$-module.

2) If $\mathfrak{R}$ is a full matrix ring over a generally noncommutative field A,

$$\mathfrak{R} = \sum c_{ik} A = \sum A c_{ik},$$

then A is reciprocally isomorphic to the automorphism field of the simple right ideals, and directly isomorphic to the automorphism field of the simple left ideals.

4. Passage from a right bimodule to a one-sided module over a product ring. The significance of the right bimodule rests essentially on this passage.

Theorem. Let $\mathfrak{M}$ be a right module over a ring $\mathfrak{T}$ which contains two elementwise commuting subrings $\mathfrak{R}$ and $\mathfrak{S}$. Then $\mathfrak{M}$ may also be viewed as a bimodule over $\mathfrak{R}$ and $\mathfrak{S}$. Conversely, if a bimodule over $\mathfrak{R}$ and $\mathfrak{S}$ is given, and if there exists a product ring $\mathfrak{T}$ in which $\mathfrak{R}$ and $\mathfrak{S}$ commute elementwise, then $\mathfrak{M}$ can also be regarded as a right module over $\mathfrak{T}$, with the $\mathfrak{T}$-operation extending the given $\mathfrak{R}$- and $\mathfrak{S}$-operations.

The product ring in the absolute automorphism ring consists of all elements of the form

$$\sum_i \rho_i\sigma_i + \rho + \sigma$$

(and if $\mathfrak{R}, \mathfrak{S}$ have identities, the extra terms ρ, σ disappear). The homomorphisms

$$\mathfrak{R} \to \overline{\mathfrak{R}}, \qquad \mathfrak{S} \to \overline{\mathfrak{S}}$$

extend by

$$\sum_i \rho_i\sigma_i + \rho + \sigma \longmapsto \sum_i \overline{\rho}_i\,\overline{\sigma}_i + \overline{\rho} + \overline{\sigma}$$

to a homomorphism $\mathfrak{T} \to \overline{\mathfrak{T}}$. Hence

$$m\Big(\sum_i \rho_i\sigma_i + \rho + \sigma\Big) = \sum_i (m\rho_i)\sigma_i + m\rho + m\sigma$$

is well-defined and defines a $\mathfrak{T}$-module containing the given $\mathfrak{R}, \mathfrak{S}$-module structure.

§2. Reciprocal and Direct Representation

1. Modules of linear forms. An $\mathfrak{S}$-right module $\mathfrak{M}$, where $\mathfrak{S}$ is a ring with identity, is called a module of linear forms in $\mathfrak{S}$ if $\mathfrak{M}$ is a direct sum of n one-term $\mathfrak{S}$-modules,

$$\mathfrak{M} = m_1\mathfrak{S} + \cdots + m_n\mathfrak{S},$$

such that $m_i\mathfrak{S}$ is operator-isomorphic to $\mathfrak{S}$; thus $m_is = 0$ always implies $s = 0$.

Theorem 1. If $\mathfrak{M}$ is a module of linear forms in $\mathfrak{S}$, then the $\mathfrak{S}$-automorphism ring $\mathfrak{U}$ of $\mathfrak{M}$ is reciprocally isomorphic, not merely homomorphic, to the ring $\overline{\mathfrak{U}}$ of all n-rowed matrices over $\mathfrak{S}$.

An arbitrary $\mathfrak{S}$-automorphism α of $\mathfrak{M}$ is completely described by

$$m_i \longmapsto \overline{m}_i = m_i\alpha;$$

hence

$$\sum_i m_is_i \longmapsto \sum_i \overline{m}_is_i = \sum_i (m_i\alpha)s_i = \sum_i (m_is_i)\alpha.$$

Writing

$$(m_1\alpha, \dots, m_n\alpha) = (m_1, \dots, m_n)A,$$

the correspondence $\alpha \mapsto A$ is one-to-one, and

$$\alpha + \beta \mapsto A + B, \qquad \alpha\beta \mapsto BA.$$

Thus the isomorphism is reciprocal. Consequently:

Theorem 1'. The automorphism ring of a module of linear forms of degree n in $\mathfrak{S}$ admits an isomorphic, faithful reciprocal representation by the full matrix ring of n-rowed matrices over $\mathfrak{S}$.

2. Representation and representation module. A module of linear forms $\mathfrak{M}$ in $\mathfrak{S}$ is called a reciprocal representation module of $\mathfrak{R}$ in $\mathfrak{S}$ if $\mathfrak{M}$ is a bimodule over $\mathfrak{R}$ and $\mathfrak{S}$ and at the same time a right module over both $\mathfrak{R}$ and $\mathfrak{S}$. It is called a direct representation module if $\mathfrak{M}$ is a bimodule and at the same time a left $\mathfrak{R}$-module and a right $\mathfrak{S}$-module.

Theorem 2. Every reciprocal, respectively direct, representation module generates a class of equivalent reciprocal, respectively direct, representations of $\mathfrak{R}$ in $\mathfrak{S}$, and all such representations are obtained in this way.

For a reciprocal representation module, $\mathfrak{R}$ maps directly homomorphically into a subring of the $\mathfrak{S}$-automorphism ring of $\mathfrak{M}$; Theorem 1' then gives a reciprocal representation. Conversely, a reciprocal representation, composed with the reciprocal isomorphism into the $\mathfrak{S}$-automorphism ring, gives a direct homomorphism from $\mathfrak{R}$, hence a representation module. The direct case is obtained by reversing the homomorphy. Direct representations of $\mathfrak{R}$ are reciprocal representations of the ring reciprocal to $\mathfrak{R}$, and conversely.

3. Passage from a reciprocal representation module to a module over an extension ring. For representation modules the transition theorem reads as follows.

If $\mathfrak{M}$ is a right module over a ring $\mathfrak{T}$ which contains two elementwise commuting subrings $\mathfrak{R}$ and $\mathfrak{S}$, and if $\mathfrak{M}$ is a module of linear forms over $\mathfrak{S}$, then $\mathfrak{M}$ may also be regarded as a reciprocal representation module of $\mathfrak{R}$ in $\mathfrak{S}$. Conversely, a reciprocal representation module may be regarded as a $\mathfrak{T}$-module whenever the product ring $\mathfrak{T}$ exists.

If $\mathfrak{M}$ is a reciprocal representation module of $\mathfrak{R}$ in $\mathfrak{S}$, and also a $\mathfrak{T}$-module with $\mathfrak{T}$ as product ring, then $\mathfrak{R}, \mathfrak{S}$-isomorphism of $\mathfrak{M}$ to a module $\mathfrak{N}$ is equivalent to $\mathfrak{T}$-isomorphism. Thus each class of isomorphic $\mathfrak{T}$-modules corresponds to a reciprocal representation class of $\mathfrak{R}$ in $\mathfrak{S}$, and conversely.

Theorem on commuting matrices. Let $\mathfrak{R} \mapsto \mathfrak{R}^*$ be a reciprocal representation of degree n of $\mathfrak{R}$ in $\mathfrak{S}$, and let $\mathfrak{B}^*$ be the ring of all n-rowed matrices over $\mathfrak{S}$ which commute elementwise with $\mathfrak{R}^*$. Then $\mathfrak{B}^*$ gives a reciprocally isomorphic representation of the $\mathfrak{R}, \mathfrak{S}$-automorphisms of the representation module generated by it, i.e. the $\mathfrak{S}$-automorphisms which are at the same time $\mathfrak{R}$-automorphisms; if the product ring $\mathfrak{T}$ exists, these are the $\mathfrak{T}$-automorphisms.

The $\mathfrak{R}, \mathfrak{S}$-automorphisms amount to a correspondence $m_i \mapsto m_i'$ which produces the same representation. The elements m_i' need not form an $\mathfrak{S}$-basis of $\mathfrak{M}$; correspondingly, the matrices in $\mathfrak{B}^*$ need not be invertible. In this extended sense

$$(m_1', m_2', \dots, m_n')a = (m_1', m_2', \dots, m_n')A$$

remains correct, although A is no longer uniquely determined by a.

The diagonal matrices $E \cdot s$ give a directly isomorphic representation of $\mathfrak{S}$, the identity representation of $\mathfrak{S}$. The correspondence from $\mathfrak{T}$ to matrices in $\mathfrak{S}$ defined by $\mathfrak{R}$ and $\mathfrak{S}$ is therefore not a representation of the ring $\mathfrak{T}$, but an extension of the reciprocal representation of $\mathfrak{R}$ to one which is operator-homomorphic over $\mathfrak{S}$:

$$\sum_i r_i s_i \longmapsto \sum_i R_i s_i.$$

In particular, the reciprocal representation of $\mathfrak{R}$ is operator-homomorphic with respect to the intersection $[\mathfrak{R}, \mathfrak{S}]$ of $\mathfrak{R}$ and $\mathfrak{S}$, which lies in the center of $\mathfrak{R}$.

§3. Modules with Respect to a Field

In what follows only representations in generally noncommutative fields will be involved. We therefore collect a few simple results on modules of linear forms over a field.

1. Normal basis of a submodule with respect to a given basis of the full module. Let A be a field and

$$\mathfrak{N} = x_1 A + \cdots + x_n A$$

a module of linear forms of rank n. Let

$$\mathfrak{L} = z_1 A + \cdots + z_\ell A$$

be a submodule of rank $\ell \le n$. The z_i are called a normal basis with respect to the x_i if they are of the form

$$z_i = x_i - (x_{\ell+1}\alpha_{i,\ell+1} + \cdots + x_n \alpha_{i,n}), \qquad i = 1, \dots, \ell.$$

Every module $\mathfrak{L}$ has, after a suitable numbering of the x_i, such a normal basis. Indeed,

$$\mathfrak{N} = \mathfrak{L} + x_{\ell+1}A + \cdots + x_nA,$$

and so

$$x_i \equiv x_{\ell+1}\alpha_{i,\ell+1} + \cdots + x_n\alpha_{i,n} \pmod{\mathfrak{L}}, \qquad i = 1, \ldots, \ell.$$

Thus the corresponding z_i lie in $\mathfrak{L}$ and exhaust $\mathfrak{L}$.

2. Extension module. Let A again be a field, and let P be a subfield. An A-module N of rank n is called an extension module of a P-module M, written $N = M_A$, if M is a submodule of N of the same rank. If

$$M = y_1P + \cdots + y_nP,$$

then

$$M_A = y_1A + \cdots + y_nA.$$

Conversely, for every P-module M, the extension module exists uniquely up to module isomorphism.

Corollary a). If $z_1, \ldots, z_s$ are linearly independent over P in M, then they are also linearly independent over A in M_A. Thus every P-module

$$T = z_1P + \cdots + z_sP$$

inside M generates an extension module

$$T_A = z_1A + \cdots + z_sA \subset M_A.$$

Corollary b). Every P-module T inside M is a contraction module, i.e. the intersection of its extension module with M:

$$T = T_A \cap M.$$

3. Theorem on invariant modules. Let $N = M_A$ be the extension module of a P-module M, and let P be the full invariant field for a group $\mathfrak{G}$ of ring automorphisms of A. The group $\mathfrak{G}$ is defined as an operator domain on N by

$$Gm = m, \qquad m \in M,$$

and

$$G\Big(\sum_i m_i\alpha_i\Big) = \sum_i m_i\, G(\alpha_i).$$

Lemma: M consists precisely of those elements of N fixed by $\mathfrak{G}$.

Theorem. The extension modules L_A of submodules L of M, and only these, are admissible submodules with respect to $\mathfrak{G}$ as operator domain.

One direction is clear. Conversely, let L be admissible for $\mathfrak{G}$, and let $z_1, \ldots, z_\ell$ be a normal basis of L with respect to a P-basis $x_1, \ldots, x_n$ of M. For every $G \in \mathfrak{G}$,

$$G(z_i) = x_i - (x_{\ell+1}G(\alpha_{i,\ell+1}) + \cdots + x_nG(\alpha_{i,n}))$$

again lies in L. Comparing coefficients in $x_1, \ldots, x_\ell$ gives $G(z_i) = z_i$. Hence, by the lemma, the z_i lie in M, and L is the extension of the P-module

$$T = z_1P + \cdots + z_\ell P$$

in M, with $T = L \cap M$.

§4. Hypercomplex Systems and Their Representation Classes

1. Extension theorem. We now combine the representation theorems of §2 with the module theorems of §3. Instead of arbitrary rings $\mathfrak{R}$, we consider hypercomplex systems with identity $S, T, \ldots$ over a commutative field P:

$$S = x_1 P + \cdots + x_n P.$$

Instead of arbitrary representation rings $\mathfrak{S}$, we consider only fields $A, B, \ldots$, unless otherwise stated with center P. In this case there always exists the product ring in which S and A commute elementwise, with intersection P; this is the direct product $S \times A$, which is at the same time the extension module S_A of S:

$$S_A = x_1 A + \cdots + x_n A.$$

It will therefore also be called the extension ring.

Under this specialization, the theorem on invariant modules becomes:

Extension theorem. Every two-sided ideal of S_A is the extension ideal of an ideal of S, namely the extension of its intersection with S.

Indeed, let the chosen group $\mathfrak{G}$ be the group of inner automorphisms of A. Then P, as center of A, is the full invariant field, while the elementwise commutativity of A with S says that $\mathfrak{G}$ is extended to S_A so that S is the full invariant domain. Every two-sided ideal $\mathfrak{a}$ of S_A is an admissible subgroup, since $\alpha^{-1}\mathfrak{a}\alpha \subseteq \mathfrak{a}$, and so it is the extension of its intersection module $\mathfrak{a} \cap S$, which is an ideal in S.

Addendum: If S is commutative, then S is the center of S_A. More generally, the center of S is also the center of S_A. By $A_r, B_n, \ldots$ we shall always mean matrix rings of the indicated degree over $A, B, \ldots$:

$$A_r = \sum A c_{ik} = \sum c_{ik} A.$$

We shall use essentially the extension theorem for simple systems:

If S is simple, then S_A is also simple:

$$S_A = \sum B c_{ik} = \sum c_{ik} B = B_t,$$

with associated field B, whose center contains P. Here A, and hence B, may have infinite rank over P; only S is assumed hypercomplex. If S and A have common center P, then P is also the center of S_A. More generally, if S is a simple hypercomplex system, then the direct product $S \times A_r$ is simple.

2. Representation classes. By a reciprocal or direct representation of a hypercomplex system we mean, as usual, a representation not equal to the zero representation and operator-homomorphic with respect to the coefficient field P.

Theorem on representation classes. Let S be hypercomplex with identity and coefficient field P, and let A be a field with center P. Then the number of distinct irreducible reciprocal representation classes of S in A is equal to the number of classes of simple right ideals in the quotient ring of S_A by its radical. If S is a simple system, then it has exactly one irreducible reciprocal and one irreducible direct representation class in A.

By the corollary to the transition theorem, the simple reciprocal representation classes and the simple module classes over S_A correspond one-to-one. Since S_A has finite rank over the field A, it satisfies the maximum and minimum conditions for one-sided ideals. If S is simple, then S_A is simple; hence there is only one simple one-sided ideal class, and so only one irreducible reciprocal representation class. The reciprocal system gives the corresponding assertion for direct representations.

3. Rank relation. If S is simple and

$$S_A = \sum B c_{ik}$$

has finite rank both over A and over B, then

$$n = rt, \qquad n = (S : P), \qquad t^2 = (S_A : B), \tag{1}$$

where r is the degree of the irreducible reciprocal representation.

4. Sharpening of the rank relation when A has finite rank over P. In this case the three relations are

$$(S:P) = n = rt, \tag{1}$$

$$(B:P)\,t = (A:P)\,r, \tag{2}$$

$$(S:P)(B:P) = (A:P)\,r^2. \tag{3}$$

Here (2) denotes the rank over P of a simple right ideal of S_A, expressed by the ranks over B and A; (3) follows from (2) by multiplying by r, using (1).

§5. Application of the Representation Theorems to Matrix Rings

The reciprocal representation of S in A considered in §4 can also be regarded as a direct homomorphism of S into a subring of A_r, where $\overline{A}$ denotes the field reciprocally isomorphic to A. The principle of the following paragraphs is the converse one: to reduce the investigation of matrix rings A_r to representation theory in the reciprocally isomorphic field A.

1. Reducible and irreducible embedding in A_r. Let A and $\overline{A}$ be reciprocally isomorphic fields, of finite or infinite rank over their center P. The simple system S over P is called irreducibly, respectively reducibly, embeddable in A_r if S has an irreducible, respectively reducible, reciprocal representation of degree r in A. If S is irreducibly embeddable in A_r, then it is reducibly embeddable in all matrix rings A_{rs}, $s > 1$. The isomorphism always extends the identity of P; by §4, 3, r divides the rank n of S over P.

In this way one obtains all simple subrings containing P, of finite rank over P, in the various A_r. For every such subring is reciprocally isomorphic to a subring of $\overline{A}_r$ and therefore has a reducible or irreducible reciprocal representation in A.

2. Theorem on inner automorphisms. Let $S^{(1)}$ and $S^{(2)}$ be two simple subrings of A_f containing P, of finite rank over P. If $S^{(1)}$ and $S^{(2)}$ are isomorphic over P, then this isomorphism is induced by an inner automorphism of A_f.

Indeed, $S^{(1)}$ and $S^{(2)}$ give generally reducible embeddings of a simple system S in A_f, hence direct representations of degree f in A. They belong to the same reducible direct representation class; the transforming matrix induces the automorphism. If A itself has finite rank over P, so that it is hypercomplex, this becomes the familiar theorem: two simple subsystems of A_f containing the center P, and isomorphic over P, are carried into one another by an inner automorphism of A_f. In particular, every automorphism of A_f is inner.

3. Theorem on elementwise commuting subrings. Let S be a simple system contained in A_f and containing P, and let R be the totality of all elements of A_f which commute elementwise with S. Then R is also a matrix ring:

$$R = B_s.$$

The associated fields B of S_A and B of R are reciprocally isomorphic. The intersection of R and S is the center of S. The ring R is a field if and only if S is irreducibly embedded in A_f.

By §2, 3, the ring R is directly isomorphic to the automorphism ring of the reciprocal representation module $\mathfrak{M}$ of S in A. This is the automorphism ring of $\mathfrak{M}$, regarded as an S_A-module. If $\mathfrak{M}$ decomposes into s simple S_A-modules and one writes, as in §4, 1,

$$S_A = \sum Bc_{ik},$$

then R is isomorphic to $\sum Bc_{ik}$, where the two fields are reciprocally isomorphic. Thus $R = B$ is a field precisely when $s = 1$, i.e. when S is irreducibly embedded. By definition, $R \cap S$ is the center of S.

4. Sharpened commutation theorem for hypercomplex matrix rings. In this case the rank relations of §4, 4 give the sharpening:

The simple subsystems of A_f which contain P, where A has finite rank over its center P, split into pairs $S, \overline{S}$, such that $\overline{S}$ consists exactly of all elements which commute elementwise with S, and conversely. The associated fields of S_A and $\overline{S}$ are reciprocally isomorphic, as are those of S and $\overline{S}_A$. The intersection of S

and $\overline{S}$ is the common center. One subring is a field if and only if the other is irreducibly embedded. The product of the ranks over P of S and $\overline{S}$ is the rank of A_f.

If $f = rs = r's'$, where S is irreducibly embeddable in A_r and $\overline{S}$ in $A_{r'}$, then the associated fields of S and $\overline{S}$ become isomorphic to a pair of commuting subrings of A_g, with $g = ss'$.

The product assertion follows directly from the rank relation. If S is irreducibly embedded, then $\overline{S}$ is a field and is reciprocally isomorphic to B, where $S_A = B_t$; the rank relation (3) of §4, 4 gives the assertion. If S is reducibly embedded, so $\overline{S} = B_s$, then $f = rs$, and multiplying (3) by s^2 gives the assertion. This also shows that $\overline{S}$ is precisely the totality of the elements commuting elementwise with S, and the remaining assertions follow from the theorem on commuting subrings.

§6. Galois Theory of Simple Systems

The sharpened commutation theorem of §5, 4 expresses the Galois theory of simple systems with respect to the center as ground domain. We consider first division rings, and then simple systems in general.

1. Galois theory of division rings of finite rank over the center. The Galois group $\mathfrak{G}$ of A is defined as the group of all automorphisms extending the identity of the center P, hence, by §5, 2, as the group of all inner automorphisms. Thus

$$\mathfrak{G} \simeq A^*/P^*,$$

where A^* and P^* denote the multiplicative groups of nonzero elements of A and P. Subgroups $\mathfrak{H}$ of $\mathfrak{G}$ therefore correspond one-to-one to subgroups H^* of A^* containing P^*, by

$$\mathfrak{H} \sim H^*/P^*.$$

A subgroup $\mathfrak{H}$ of $\mathfrak{G}$ is called closed if, after adjoining zero, H^* becomes a division ring H. The fact that C admits the group $\mathfrak{H}$ is equivalent to saying that C commutes elementwise with H. Hence the commutation theorem of §5, 4 becomes the following.

Main theorem of the Galois theory of noncommutative division rings. The division rings C between P and A and the closed subgroups $\mathfrak{H}$ of $\mathfrak{G}$ correspond bijectively in such a way that C is the full invariant division ring of $\mathfrak{H}$, and $\mathfrak{H}$ is the full invariant group of C.

2. Galois theory of simple systems. If A_f is a simple system with center P, the Galois group $\mathfrak{G}$ is again the group of inner automorphisms, hence isomorphic to

$$A_f^*/P^*,$$

where now A_f^* denotes the multiplicative group of the regular elements of A_f. A subgroup $\mathfrak{H} \sim H^*/P^*$ of $\mathfrak{G}$ is called simple-closed if the subring H generated by H^* is a simple system and if H^* is the set of all regular elements of H. The passage from the commutation theorem to Galois theory is supplied by the following lemma.

Lemma. Every simple system in A_f containing P is generated by the group H^* of its regular elements. This is clear if P has infinitely many elements: the "general element" formed with indeterminates is regular, and by suitable specialization one obtains regular basis elements over P. The lemma holds in general by Shoda.[1]

By the lemma, commutation of S with a simple system $\mathfrak{S}$ is again equivalent to saying that S admits the automorphisms induced by the regular elements of $\mathfrak{S}$. Thus the commutation theorem of §5, 4 becomes also here the following.

Main theorem of the Galois theory of simple systems. The simple systems S in A_f containing P and the simple-closed subgroups $\mathfrak{H}$ of $\mathfrak{G}$ correspond bijectively in such a way that S is the full invariant domain of $\mathfrak{H}$, and $\mathfrak{H}$ is the full invariant group of S.

3. Proof by means of the extension principle. For the case of a division ring I give a second proof of the main theorem. It rests on the principle of extending isomorphisms, i.e. representations of first degree, and since it uses no count of ranks it makes fewer finiteness assumptions. In particular it shows in what direction a transfer to division rings S of infinite rank will be possible. The proof also does not use the fact that there is only one irreducible reciprocal representation class; it therefore applies exactly in the same way in the commutative case, §8, 2. I first state some lemmas.

[1] Shoda, work cited in note 3); there the introduction of indeterminates is avoided.

Hypotheses. The simple system S over P is to admit a reciprocal representation of first degree in A, and is therefore a division ring. Here A may have finite or infinite rank over its center P. A decomposition of S_A into n simple operator-isomorphic right ideals, as in §4, 3, is given by

$$S_A = r_1 + \cdots + r_n = e_1 S_A + \cdots + e_n S_A = e_1 A + \cdots + e_n A;$$

the reciprocal representation generated by e_i is therefore defined by

$$e_i \sigma = e_i \sigma_i.$$

Lemma 1. The n reciprocal representations generated by $e_1, \ldots, e_n$ are distinct, hence are distinct representations of the same representation class. Thus S has at least as many distinct representations as its rank.

To prove this, we extend the representations from S, as at the end of §2, to correspondences of S_A which are operator-homomorphic over A. Here these correspondences are defined by

$$e_i w = e_i \omega$$

with $\omega \in A$, for each $w \in S_A$. If e_i and e_j, $i \neq j$, generated the same representation, one would have $e_i w = e_j w$ for every $w \in S_A$. This is impossible because $e_i e_i = e_i$ and $e_j e_i = 0$.

Lemma 2. Let T be a division ring between P and S, and let s be the rank of S over T. Then every reciprocal isomorphism of T into A admits at least s different extensions.

Let the reciprocal isomorphism, i.e. the reciprocal representation of T in A, be mediated by a decomposition

$$T_A = E_1 T_A + \cdots + E_h T_A = E_1 A + \cdots + E_h A, \qquad E_i t = E_i \tau,$$

where the E_i are idempotents. This is no restriction, since a representation generated by a basis element $E_i \alpha$ is also generated by $\alpha^{-1} E_i \alpha$, hence by an idempotent. Right multiplication by S gives

$$S_A = E_1 S_A + \cdots + E_h S_A.$$

The $E_i S_A$ all have rank s over A. Indeed the rank of $E_i S_A$ is at most s, as follows by substituting a T-basis into S, using the commutativity of S and A:

$$S = T w_1 + \cdots + T w_s,$$

$$E_i S_A = E_i T_A w_1 + \cdots + E_i T_A w_s = E_i w_1 A + \cdots + E_i w_s A.$$

Since the sum S_A has rank $n = hs$, every $E_i S_A$ has exactly rank s. Thus

$$E_i S_A = r_{i1} + \cdots + r_{is} = e_{i1} A + \cdots + e_{is} A,$$

where the s representations generated by $e_{i1}, \ldots, e_{is}$ are all distinct by Lemma 1. They are all extensions of the representation of T generated by E_i, since $E_i t = E_i \tau$ implies

$$(e_{i1} + \cdots + e_{is}) t = (e_{i1} + \cdots + e_{is}) \tau$$

and hence $e_{ij} t = e_{ij} \tau$, by the direct decomposition.

Definition. The partition $\mathfrak{I}_T$ of the reciprocal isomorphisms $\mathfrak{I}$ from S into A, induced by T, is defined as follows: precisely those isomorphisms in $\mathfrak{I}$ are regarded as equivalent which induce the same isomorphism on T.

Main theorem. If $\mathfrak{I}_T$ is the partition induced by T, then T is a maximal subfield relative to this partition.

For if Z is an intermediate division ring between T and S, of degree l over T, then by Lemma 2 every class of $\mathfrak{I}_T$ splits into at least l classes of $\mathfrak{I}_Z$. The transition from this form of the theorem to the usual one arises by multiplying the elements of the set $\mathfrak{I}$ by the inverse of any fixed one of them; the set then becomes the automorphism group, and a class in $\mathfrak{I}_T$ becomes the invariant group of T. The statement that closed subgroups are full invariant groups is included in this: for $\mathfrak{H} \sim H^*/P^*$, one simply puts the division ring H in place of the intermediate division ring T.

§7. The Class of Similar Algebras. Splitting Fields

From now on only division rings $A, \overline{A}$ and matrix rings A_f of finite rank over their center P will occur. Thus they are simple normal algebras over P, and will be called algebras over P, or simply algebras; $A, \overline{A}$ are then division algebras. All A_f with the same associated A are collected into one class of similar algebras A, A_f. Isomorphic A's are identified.

1. The group of algebra classes. If A, B are algebras over P, then the same is true of their product, the direct product over P; for by the end of §3,

$$A_f \times_P B_g = C_h,$$

where C is a uniquely determined, up to isomorphism, division algebra with center P. Multiplication of A_f, B_g by matrix rings over P shows that this multiplication is uniquely determined for the classes, which therefore form a multiplicatively closed, commutative system. Moreover R. Brauer's theorem holds: the classes of similar algebras form an abelian group under direct product. The system has an identity class, consisting of matrix rings over P, that is, of the algebras similar to P. Also each class (A) has the inverse $(\overline{A}) = (A)^{-1}$, consisting of the algebras similar to $\overline{A}$, where $\overline{A}$ is reciprocally isomorphic to A. This follows from the commutation theorem §5, 3 by the specialization $S = A$. For A can be embedded irreducibly in itself; one obtains $B = P$ and $A_A = P_f$.[2]

2. Splitting fields of an algebra class. The theory of splitting fields again rests entirely on the principle stated at the beginning of §5: the commutative extension fields are represented in A or in $\overline{A}$. We first need the following lemma.

Lemma. If A is a division ring with center P, and Z is a finite commutative extension field of P, then A_Z is simple with center Z. For by §4, 1 this holds for the system $Z_{\overline{A}}$, reciprocally isomorphic to A_Z. Thus for every class (A) of algebras over P similar to A, the field Z gives an extension class $(A)_Z$ of simple algebras over Z similar to A_Z.

The associated division algebra D of an extension class follows immediately from the commutation theorem of §5, 3.

Theorem on the associated division algebra of an extension class. Let $(A)_Z$ be an extension class, and let Z denote an irreducible embedding, i.e. representation, of Z in A_f. Then

$$(A)_Z = (D),$$

where D is the set of all elements of A_f which commute elementwise with Z. One only has to replace the simple system S in §5, 3 by Z, observing that Z_A and A_Z are reciprocally isomorphic. If Z is embedded reducibly, the totality of elements commuting elementwise with Z is a matrix ring over D.

A commutative extension field Z of P is called a splitting field of the class (A) if the extension class $(A)_Z$ splits completely, that is, is equal to the identity class over Z, the class of all matrix rings over Z.

Theorem on the characterization of splitting fields. A finite commutative extension field Z of P is a splitting field of the class (A) if and only if its irreducible embedding in A_f gives a maximal commutative subfield of A_f.

Indeed, the property that Z be a splitting field of (A) is equivalent to $(D) = (Z)$. Hence, by the theorem on the associated division algebra, the irreducible embedding Z must be a maximal commutative subfield. If this condition is fulfilled, then necessarily $D = Z$, since adjoining any $a \in D$ to Z gives a commutative extension field.

Remarks. 1. It follows at the same time that a maximal commutative subfield Z, when irreducibly embedded, generates a maximal commutative subring. For the elements commuting elementwise with Z form a division ring.

2. The theorem also gives the existence of splitting fields, since the associated division algebra A certainly has maximal commutative subfields.

[2]The finer theorems on the group of algebra classes use the theory of factor systems; that will not be taken up here. Notice that up to this point no considerations concerning commutative fields have occurred. For example, the fact that the rank $(A : P)$ is a square has neither been used nor proved.

3. Rank relations, index, infinite commutative extensions. The rank statement from §5, 4 first gives the known fact that the rank of a simple algebra over its center is a square. For if Z is maximal commutative in A, then, since every embedding in A is irreducible,

$$(Z : P)(Z : P) = (A : P),$$

hence

$$(A : P) = m^2, \qquad (Z : P) = m,$$

where m is Schur's index; it is also the degree of all splitting fields embeddable in the division algebra A itself. For the degree n of a general splitting field one obtains

$$n = mr,$$

for instance from $(A_r : P) = m^2 r^2$, or also from the rank relation §4, 3, since the index m is also the number of simple components of A_Z, which becomes a matrix ring over Z of rank m^2. More generally, if $A_Z \simeq D_l$, if L is the irreducible embedding of A in A_r, and if d^2 is the rank $(D : L)$ of the division algebra D over its center Z, then

$$l\,d = mr.$$

Indeed, by §5, 4 and the theorem on the associated division algebra from 2.,

$$(A_r : P) = (L : P)(D : P),$$

so

$$m^2 r^2 = l^2 d^2.$$

From the existence of splitting fields follow the facts for infinite commutative, algebraic or transcendental, extension fields Ω of P: the extension class $(A)_\Omega$ is a class of simple algebras with center Ω. If in particular Ω is algebraically closed, then A_Ω is a matrix ring of degree m. Thus the index is equal to the "absolute number of components." For if Z is a splitting field of A and Ω' is the compositum of Ω and Z, then $A_{\Omega'}$ is a matrix ring over Ω', hence has no radical and has center Ω'. Therefore A_Ω also has no radical and its center is Ω; any element in the center of A_Ω not belonging to Ω would also lie in the center of $A_{\Omega'}$. Hence A_Ω is a simple algebra over Ω, and certainly a matrix ring over Ω if Ω is algebraically closed.[3]

4. Existence of separable splitting fields. Every class (A) has separable splitting fields, indeed such that can be embedded in A itself.[4]

Let the ground field P have characteristic p, and let the index be

$$m = sp^f$$

with s prime to p. If Z is a splitting field of A of degree m, and Z_1 is the extension of first kind contained in it, so that

$$(Z : Z_1) = p^f,$$

then the index of $A_{Z_1} \sim D$ is p^f. We prove the existence of a separable extension field Z_2 of Z_1 such that the index of D_{Z_2} is a proper divisor of p^f. Now D_Ω, with Ω algebraically closed, has no radical by 3.; the reduced discriminant of D therefore does not vanish. Hence there is at least one element $d \in D$ with nonzero reduced trace. This d cannot already lie in the ground field Z_1, because the trace of every element $\alpha \in Z_1$ is $p^f\alpha$, hence vanishes. Thus $Z_2 = Z_1(d)$ is a proper, indeed separable, extension field; the index of D_{Z_2} is a proper divisor of p^f. Repeating this finitely many times gives a separable splitting field of (A) whose degree m agrees with the index of A.

§8. Splitting Fields, Decomposition Fields, and Galois Theory in the Commutative Case

In order to pass from the splitting fields of systems simple over their center to those of arbitrary simple systems, one still has to develop the splitting theory of the center; a Galois theory parallel to §6, 3 is to be added to it. All fields and systems occurring in this paragraph are commutative.

[3] In general there are also "minimal" splitting fields, i.e. such that no proper subfield is a splitting field, of all possible degrees. Compare Brauer–Noether, the work cited in note 2).

[4] Compare G. Köthe, Über Schiefkörper mit Unterkörpern zweiter Art über dem Zentrum, Journ. f. Math. 166 (1932), pp. 182–184. The present, much simpler proof goes back to a remark of M. Zorn.

1. Splitting and decomposition fields of commutative simple systems. Let the commutative system Z be simple over P, hence a field, and let Ω be an algebraically closed extension field of P. It is known that Z_Ω remains a system without radical if and only if Z is separable, i.e. an extension of first kind, over P. For only then does it have as many isomorphic maps of first degree into Ω as its rank over P, hence as many distinct representations, which in this case coincide with representation classes.

The possibility that a radical occurs in Z_Ω, so that a direct decomposition into absolutely simple components need not take place, leads to the following definitions. An extension field Λ of P is called a splitting field if Z_Λ splits into composition factors of first degree; in other words, if all absolutely irreducible representations already lie in Λ. An extension field T of P is called a decomposition field if Z_T splits off at least one composition factor of first degree; in other words, if at least one absolutely irreducible representation of Z exists in T.

Splitting and decomposition fields are characterized by the following theorem.

Theorem. A field T is a decomposition field if and only if it contains a subfield Z' isomorphic to Z. A field Λ is a splitting field if and only if it contains a subfield Γ isomorphic to the Galois field belonging to Z.

This follows directly from the definitions of splitting and decomposition fields in terms of absolutely irreducible representations. In particular, in analogy with the noncommutative case, one obtains the following corollary: a field Z' is a minimal decomposition field, i.e. no proper subfield is a decomposition field, if and only if it is isomorphic to Z. A minimal decomposition field is at the same time a splitting field if and only if Z is normal, hence a Galois field over P.

This is the reason for calling a simple algebra normal over its center. In a fixed algebraically closed Ω over P, there is only one minimal splitting field, namely the Galois field belonging to Z, in contrast to the generally infinitely many nonisomorphic fields in the noncommutative case. This has its source in the uniqueness of the direct decomposition in the commutative case, in contrast to the noncommutative case where uniqueness is only up to operator isomorphism; equivalently, in the finitely many different absolutely irreducible representations in the commutative case, as opposed to the infinitely many different ones in the noncommutative case, although they belong to the same class.

2. Hypercomplex proof of Galois theory in the case of separable fields Z/P. In exact analogy with §6, 3, the theory of isomorphisms holds for arbitrary Z separable over P; if Z is Galois, one can then pass to the usual automorphism group.

Hypotheses. The simple system Z over P is a finite separable extension field, Ω denotes a finite or infinite splitting field over P, $Z' = Z^{(1)}$ a subfield isomorphic to Z, and Γ the corresponding Galois field. By 1. there is a direct decomposition

$$Z_\Omega = r^{(1)} + \cdots + r^{(n)} = e^{(1)} Z_\Omega + \cdots + e^{(n)} Z_\Omega = e^{(1)}\Omega + \cdots + e^{(n)}\Omega.$$

The representation $Z \to Z^{(i)}$ generated by $e^{(i)}$, with $Z^{(i)} \subseteq \Gamma$, is defined by

$$e^{(i)} z = e^{(i)} z^{(i)}.$$

Lemma 1. The n representations generated by $e^{(1)}, \ldots, e^{(n)}$ are all distinct. The proof is as in §6, 3, or also follows from 1.; they belong to distinct representation classes.

Lemma 2. If T is an intermediate field between P and Z, and s is the rank of Z over T, then every isomorphism of T into Ω admits at least, and hence exactly, s extensions.

The proof is as in §6, 3. The strengthening of "at least" to "exactly" follows from the uniqueness of the decomposition of Z_Ω, according to which Z has not merely at least but exactly n isomorphisms in Ω.

As in §6, 3, one defines the partition $\mathfrak{I}_T$ induced by T of the finite set $\mathfrak{I}$ of isomorphisms of Z: precisely those isomorphisms in $\mathfrak{I}$ are regarded as equivalent in $\mathfrak{I}_T$ which induce the same isomorphism on T. One proves the following.

Main theorem. If $\mathfrak{I}_T$ is the partition induced by T, then T is a maximal subfield with respect to this partition.

From here, in the case of a Galois Z, one passes to the automorphism group and to the usual formulation exactly as in §6, 3 by composing with the reciprocal of an isomorphism. The corresponding assertion for subgroups follows from the consideration of idempotents, which is of interest in itself.

3. The idempotents of Z_Ω. Let S run through the n substitutions which carry Z into the individual $Z^{(i)}$, that is, the compositions of the isomorphisms $Z \to Z_1$ and $Z \to Z^{(i)}$, where the first is the reciprocal of $Z \to Z_1$. The field Z need not be Galois. For elements $a \in Z_\Omega$, the substitutions S are defined as operators by stipulating that they induce the identity on Z. Thus for each a the conjugate a^S lying in $Z_{(i)}$ is defined. With these conventions one has the following theorem.

Theorem. The n idempotents $e^{(i)}$ of Z_Ω are conjugate:

$$e^{(i)} = e^S, \qquad e = e^{(1)};$$

they lie in the n conjugate extension systems $Z\Omega$.

Indeed, applying S carries the defining relations

$$ez = ez^{(1)}, \qquad e^2 = e$$

to

$$e^S z = e^S z^{(i)}, \qquad (e^S)^2 = e^S.$$

By uniqueness of the decomposition the idempotents are themselves uniquely determined; hence $e^S = e^{(i)}$. Since Z is also a decomposition field, so that the representation $ez = ez^{(1)}$ is already mediated, e already lies in $Z\Omega$, and consequently e^S lies in $Z^S\Omega$.

If Z is Galois, the part of the main theorem of Galois theory concerning subgroups follows immediately. If $\mathfrak{H}$ is a subgroup of the Galois group $\mathfrak{G}$, then, by the uniqueness of the components of a direct sum, the element

$$\sum_{H \in \mathfrak{H}} e^H$$

admits all substitutions from $\mathfrak{H}$ and no others. Since the group $\mathfrak{G}$ was defined for Z, this is the Galois theory of Z; by isomorphism it also settles the theory for Z'.

4. Connection of the idempotents with complementary bases. Again Z is not assumed to be Galois.

Theorem. Let $a_1, \ldots, a_n$ be a basis of Z over P, and write the idempotent as

$$e = a_1\beta_1 + \cdots + a_n\beta_n,$$

so that

$$e^S = a_1\beta_1^S + \cdots + a_n\beta_n^S.$$

Then

$$\beta_1^S, \ldots, \beta_n^S$$

are the images, under the map mediated by e^S, of the complementary basis $b_1, \ldots, b_n$ to $a_1, \ldots, a_n$.

For the map mediated by e^S, $e^S x = e^S x^S$, in particular $e^S a_i = e^S a_i^S$, the relations

$$e^S e^S = e^S, \qquad e^S e^R = 0 \quad (S \neq R)$$

give the assignments $e^S \mapsto 1$, $e^R \mapsto 0$. Hence

$$1 = a_1^S\beta_1^S + \cdots + a_n^S\beta_n^S, \qquad 0 = a_1^R\beta_1^S + \cdots + a_n^R\beta_n^S \quad (S \neq R).$$

Written in matrix form, this is the defining equation for the complementary bases:

$$\begin{pmatrix} a_1^1 & \cdots & a_n^1 \\ \vdots & \ddots & \vdots \\ a_1^n & \cdots & a_n^n \end{pmatrix} \begin{pmatrix} \beta_1^S \\ \vdots \\ \beta_n^S \end{pmatrix} = \begin{pmatrix} 0 \\ \vdots \\ 1 \\ \vdots \\ 0 \end{pmatrix}.$$

The passage to the trace definition follows because, for

$$a_i = \sum_S a_i^S e^S, \qquad b_i = \sum_S b_i^S e^S,$$

the matrix relation, after interchanging rows, becomes

$$\delta_{ij} = \mathrm{Sp}(a_i b_j).$$

Here the a_i and b_i lie in Z, since they admit all substitutions S.[5,6]

The contragredience of the complementary bases follows here from the fact that the idempotents e^S are invariants of Z, independent of the basis chosen. Thus all

$$a_1\beta_1^S + \cdots + a_n\beta_n^S$$

are transformed into themselves when one passes to another basis

$$a_1', \ldots, a_n'$$

of Z/P.

§9. Splitting Fields and Decomposition Fields of Arbitrary Systems

1. Definitions and reduction to the simple case. The definitions given in §8 correspond in the general case to the following. Let S be hypercomplex over P. A commutative extension field Λ of P is called a splitting field of S if, in a composition series by one-sided ideals, S_Λ splits into absolutely simple composition factors; in other words, if all irreducible representations of S in Λ are already absolutely irreducible. An extension field T of P is called a decomposition field if S_T splits off at least one absolutely simple composition factor; equivalently, if at least one representation irreducible in T is already absolutely irreducible.

These definitions plainly contain the ones given in §8 for the commutative case and in §7 for simple algebras. The latter is true because A_Λ, for every commutative extension of the center, is completely reducible, so that composition factors and components of the direct sum decomposition coincide. Full matrix rings over the center, and only these, give absolutely simple components, hence also absolutely irreducible representations. Moreover A_Λ is two-sided simple, hence decomposes into operator-isomorphic components. Therefore the splitting off of one absolutely simple component already gives complete splitting: decomposition fields and splitting fields coincide.

From the definitions the following facts follow immediately: the splitting fields of S are the composita obtained by taking one splitting field for each representation irreducible in P; a decomposition field of S is any decomposition field of a representation irreducible in P. Because of the one-to-one correspondence between the simple systems, namely the components of the residue-class ring modulo the radical, and the representations irreducible in P, it is enough to consider simple systems. The results will then follow by combining §7 and §8.

2. Splitting fields and decomposition fields of simple systems. We first consider the case of a simple system A with separable center Z over P. From §8, 1 and §8, 3 one obtains the following. The two-sided decomposition of A_Ω corresponding to the direct decomposition of Z_Ω, where Ω denotes a splitting field of Z, gives n conjugate components

$$e^S A_\Omega$$

of A_Ω, each isomorphic to A, which become simple algebras over their centers

$$e^S Z_\Omega = e^S Z.$$

The substitutions S are defined as operators for A_Ω by stipulating that they induce the identity on A.

Indeed, by §8, 3 the idempotents are conjugate; the same is therefore true of the components $e^S A_\Omega$, by the definition of the substitutions for A. Since A is two-sided simple, $e^S A_\Omega$ is ring-isomorphic to A.

[5] Compare Representation Theory §25.

[6] Connections between complementary bases and idempotents occur first in Dedekind, Zur Theorie der aus n Haupteinheiten gebildeten komplexen Größen; compare vol. II of the collected works, pp. 4–5.

Moreover $e^S A_\Omega = e^S A_{Z^S}$, because $e^S Z_\Omega = e^S Z$; hence the center coincides with the coefficient domain, and the components are normal algebras.

The results of §7, 2 now further imply: if A is a simple system over P with separable center Z over P, then A_Ω too is without radical, hence completely reducible; here Ω may be any finite or infinite extension field. For by §7, 2, $e^S A_\Omega$ remains simple after every extension of coefficients, and therefore has no radical. Hence the same holds for the direct sum; this is A_Ω, or, if Ω is not a splitting field of Z, is obtained by coefficient extension from A_Ω.

From §7, 2 follows also the following characterization.

Characterization of decomposition fields and splitting fields. If A is a division algebra with separable center Z, then the irreducible embeddings of the decomposition fields are given by all and only the maximal commutative subfields of the A_f containing Z. The splitting fields are all and only the composita of a decomposition field with a splitting field of the center, in particular with the Galois field belonging to the center. A minimal decomposition field may be a splitting field even when the center is not Galois.

A decomposition field must contain a field isomorphic to Z; the assertion about embeddings of decomposition fields now follows from the preceding facts and from §7, 2. A splitting field must moreover contain a splitting field Ω of Z. But every decomposition field over Ω is at the same time a splitting field, since the components $e^S A_\Omega$ are simple and have a center isomorphic to Ω. If Z is Galois, minimal decomposition fields and splitting fields therefore coincide. But even in the non-Galois case, unlike the commutative case, there may be minimal decomposition fields which are at the same time splitting fields, namely whenever such a minimal decomposition field contains the Galois field belonging to Z. Example: a quaternion division algebra with center isomorphic to $P(\sqrt{2})$, P the field of rational numbers; the Galois field belonging to $P(\sqrt{2})$ is a minimal decomposition and splitting field.

Finally, if the center is inseparable, then Z_Ω, and with it A_Ω, is a system with radical. Let $\mathfrak{C}$ be the radical of Z_Ω, and let $\mathfrak{C}A_\Omega$ be the extension ideal in A_Ω. Then $Z_\Omega/\mathfrak{C}$ has a direct decomposition into conjugate components, in exact analogy with §8, 2. This gives, just as above, the direct decomposition of $A_\Omega/\mathfrak{C}A_\Omega$, in such a way that the components $e^S A_\Omega$ are isomorphic to A, with center $e^S Z^\Omega$. Thus the theorems on decomposition and splitting fields remain valid. The count of ranks further shows that the composition factors, corresponding to a composition series of $\mathfrak{C}A_\Omega$, are operator-isomorphic to $A_\Omega/\mathfrak{C}A_\Omega$, not merely homomorphic.

Expressed for irreducible representations, this gives the summary: if a representation of a hypercomplex system irreducible over P is given, then the representation is absolutely completely reducible if and only if the associated center is separable over P. In every case, the splitting and decomposition fields of the representation can be characterized as above by embedding into the associated simple system. In particular, the lowest degree of a decomposition field is the product of the degree of the center and the index of the associated algebra class over the center; the degree of every decomposition field is a multiple of this. The representation decomposes, in the sense of composition series, into as many distinct but conjugate absolutely irreducible representation classes as the degree of the largest separable field contained in the center. Each class occurs as many times as is given by the product of the index and the degree of the center over this separable extension.

For a perfect ground field, where only separable extension fields occur, this returns the known results of I. Schur, with the addition of the embedding characterization which already occurs in R. Brauer.

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